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Course: Fundamentals of data science

Course code: DSA0404

CO3-AT1 Case Study

1.) Detecting an Unusual Customer

A shopping website finds that the average amount spent by customers is ₹2,000, with a

standard deviation of ₹400.

A particular customer spends ₹3,200.

1. Calculate the customer's z-score.

2. Is the customer's spending unusually high compared with the average?

3. What would a negative z-score mean in this context?

4. If another customer has a z-score of −1.5, what does it tell you?

A.) Given:

- Mean (μ) = ₹2,000
- Standard Deviation (σ) = ₹400
- Customer Spending (x) = ₹3,200

Step 1: Write the Formula

$$z = \frac{x - \mu}{\sigma}$$

where:

- x= Customer spending
- μ= Mean spending
- σ= Standard deviation

Step 2: Substitute the Values

$$z = \frac{3200 - 2000}{400}$$

Step 3: Subtract the Mean from the Customer Spending

$$3200 - 2000 = 1200$$

So,

$$z = \frac{1200}{400}$$

Step 4: Divide

$$z = \frac{1200}{400} = 3$$

Z=3

2.) Delivery Time Analysis

A delivery company records delivery times (in minutes):

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

1. Calculate the mean and median.

2. Find Q1 and Q3.

3. Calculate the IQR.

4. Check for outliers using the 1.5 × IQR rule.

5. Explain whether the mean or median gives a better description of the typical delivery time.

A.) Given

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

1. Calculate the Mean

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Step 2: Find the sum

$$25 + 27 + 28 + 29 + 30 + 30 + 31 + 32 + 34 + 60 = 326$$

Step 3: Count the values

$$n = 10$$

Step 4: Calculate the mean

$$\text{Mean} = \frac{326}{10} = 32.6$$

Mean = 32.6 minutes

2. Calculate the Median

Step 1: Arrange the data (already arranged)

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

Step 2: Number of values = 10 (Even)

Median is the average of the 5th and 6th values.

$$\text{Median} = \frac{30 + 30}{2} = 30$$

Median= 30 minutes

3. Find Q1 and Q3

Step 1: Lower half

25, 27, 28, 29, 30

Middle value = 28

$Q_1 = 28$

Step 2: Upper half

30, 31, 32, 34, 60

Middle value = 32

$Q_3 = 32$

Answer:

- Q1 = 28
- Q3 = 32

4. Calculate the IQR

Step 1: Formula

$IQR = Q_3 - Q_1$

Step 2: Substitute

$IQR = 32 - 28 = 4$

IQR = 4

5. Check for Outliers (1.5 × IQR Rule)

Step 1: Calculate 1.5 × IQR

$1.5 \times 4 = 6$

Step 2: Lower Limit

$28 - 6 = 22$

Step 3: Upper Limit

$32 + 6 = 38$

Step 4: Compare the data

Values below 22 → None

Values above 38 → 60

❖ 60 is an outlier.

3.) Two Cities' Temperatures

The average daily temperatures (°C) for 7 days are:

City A: 28, 29, 30, 30, 31, 30, 32

City B: 20, 25, 30, 35, 30, 25, 45

1. Calculate the mean temperature for each city.

2. Calculate the variance for each city.

3. Which city has greater temperature variability?

4. Why can two datasets have the same mean but different standard deviations?

5. Which city has more predictable temperatures?

A.) Given

City A: 28, 29, 30, 30, 31, 30, 32

City B: 20, 25, 30, 35, 30, 25, 45

1. Calculate the Mean Temperature

Formula

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

City A

Step 1: Find the sum

$$28 + 29 + 30 + 30 + 31 + 30 + 32 = 210$$

Step 2: Count the values

$$n = 7$$

Step 3: Calculate the mean

$$\text{Mean} = \frac{210}{7} = 30$$

Mean of City A = 30°C

City B

Step 1: Find the sum

$$20 + 25 + 30 + 35 + 30 + 25 + 45 = 210$$

Step 2: Count the values

$$n = 7$$

Step 3: Calculate the mean

$$\text{Mean} = \frac{210}{7} = 30$$

Mean of City B = 30°C

2. Calculate the Variance

Formula

$$\text{Variance} = \frac{\sum(x - \bar{x})^2}{n}$$

where $\bar{x} = 30$

City A

x	x − 30	(x − 30)²
28	-2	4
29	-1	1
30	0	0
30	0	0
31	1	1
30	0	0
32	2	4

Step 1: Sum of squared differences

$$4 + 1 + 0 + 0 + 1 + 0 + 4 = 10$$

Step 2: Divide by 7

$$\frac{10}{7} = 1.43$$

Variance of City A = 1.43

City B

x	$x - 30$	$(x - 30)^2$
20	-10	100
25	-5	25
30	0	0
35	5	25
30	0	0
25	-5	25
45	15	225

Step 1: Sum of squared differences

$$100 + 25 + 0 + 25 + 0 + 25 + 225 = 400$$

Step 2: Divide by 7

$$\frac{400}{7} = 57.14$$

Variance of City B = 57.14

3. Which City Has Greater Temperature Variability?

Compare the variances:

- City A = 1.43
- City B = 57.14

❖ Since 57.14 > 1.43, City B has greater temperature variability.

4.) Data Center CPU Usage

A data scientist records CPU usage percentages:

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

1. Calculate mean, median, and mode.

2. Calculate the range.

3. Find Q1, Q3 and IQR.

4. Identify possible outliers.

5. Determine the direction of skewness.

A.) Given

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

1. Calculate the Mean

Step 1: Formula

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Step 2: Find the sum

$$40 + 42 + 45 + 45 + 46 + 48 + 50 + 52 + 55 + 90 = 513$$

Step 3: Count the values

$$n = 10$$

Step 4: Calculate the mean

$$\text{Mean} = \frac{513}{10} = 51.3$$

Mean = 51.3%

2. Calculate the Median

Step 1: Data is already arranged

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

Step 2: Number of values = 10 (Even)

Median is the average of the 5th and 6th values.

$$\text{Median} = \frac{46 + 48}{2} = 47$$

Median = 47%

3. Calculate the Mode

The mode is the value that occurs most often.

45 appears 2 times.

All other values appear once.

Mode = 45%

4. Calculate the Range

Formula

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Calculation

$$90 - 40 = 50$$

Range = 50%

5. Find Q1, Q3 and IQR

Lower Half

40, 42, 45, 45, 46

Middle value = 45

$$Q_1 = 45$$

Upper Half

48, 50, 52, 55, 90

Middle value = 52

$Q_3 = 52$

Calculate IQR

$IQR = Q_3 - Q_1$
$IQR = 52 - 45 = 7$

- Q1 = 45
- Q3 = 52
- IQR = 7
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6. Identify Possible Outliers

Step 1: Calculate 1.5 × IQR

$1.5 \times 7 = 10.5$

Step 2: Lower Limit

$45 - 10.5 = 34.5$

Step 3: Upper Limit

$52 + 10.5 = 62.5$

Step 4: Compare the data

- Values below 34.5 → None
- Values above 62.5 → 90

❖ 90 is an outlier.

5.) Comparing Two Datasets

Two data analysts receive the following datasets:

Dataset A: 10, 20, 30, 40, 50

Dataset B: 28, 29, 30, 31, 32

1. Calculate the mean of both datasets.

2. Calculate the variance of both.

3. Calculate the standard deviation of both.

4. Which dataset has greater spread?

5. If both datasets represent model errors, which model appears more consistent?

6. Explain why mean alone is not enough to compare datasets.

A.) Given Data:

- Dataset A: 10, 20, 30, 40, 50
- Dataset B: 28, 29, 30, 31, 32

1. Calculate the Mean of Both Datasets

Formula

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Dataset A

Step 1: Find the sum

$$10 + 20 + 30 + 40 + 50 = 150$$

Step 2: Count the values

$$n = 5$$

Step 3: Calculate the mean

$$\frac{150}{5} = 30$$

Mean of Dataset A = 30

Dataset B

Step 1: Find the sum

$28 + 29 + 30 + 31 + 32 = 150$

Step 2: Count the values

$n = 5$

Step 3: Calculate the mean

$$\frac{150}{5} = 30$$

Mean of Dataset B = 30

2. Calculate the Variance

Formula

$$\text{Variance} = \frac{\sum(x - \bar{x})^2}{n}$$

where Mean = 30

Dataset A

x	x − 30	(x − 30)²
10	-20	400
20	-10	100

x	$x - 30$	$(x - 30)^2$
30	0	0
40	10	100
50	20	400

Step 1: Sum of squared differences

$$400 + 100 + 0 + 100 + 400 = 1000$$

Step 2: Divide by 5

$$\frac{1000}{5} = 200$$

Variance of Dataset A = 200

Dataset B

x	$x - 30$	$(x - 30)^2$
28	-2	4
29	-1	1
30	0	0
31	1	1
32	2	4

Step 1: Sum of squared differences

$$4 + 1 + 0 + 1 + 4 = 10$$

Step 2: Divide by 5

$$\frac{10}{5} = 2$$

Variance of Dataset B = 2

3. Calculate the Standard Deviation

Formula

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

Dataset A

$$\sqrt{200} = 14.14$$

Standard Deviation = 14.14

Dataset B

$$\sqrt{2} = 1.41$$

Standard Deviation = 1.41

4. Which Dataset Has Greater Spread?

Compare the standard deviations:

- Dataset A = 14.14
- Dataset B = 1.41

❖ Since 14.14 > 1.41, Dataset A has the greater spread.